

Time: Three Hours

Maximum Marks: 75

SECTION– A ($10 \times 2 = 20$ marks)

Answer any TEN questions

1. Find the value of $I_n = \int_0^{\frac{\pi}{2}} \sin^4 x \, dx$
2. Find $\int_0^{\frac{\pi}{2}} \sin^6 x \cos^5 x \, dx$
3. Write about the Even and Odd functions
4. Find the constant term in the Fourier series $f(x) = \cos^2 x$ in interval $(-\pi, \pi)$.
5. Solve $p - \frac{1}{p} = \frac{x}{y} - \frac{y}{x}$.
6. Evaluate $p^2 + pxy = y^2 \log y$.
7. Solve $p^5 + py = 1$.
8. Solve $L(\sin at)$.
9. Find the Laplace transforms of $(1 + 2t^2)^2$.
10. Define Vector differential operator.
11. Prove that $\mathbf{w} \times \mathbf{r}$ is solenoidal if $\mathbf{w}$ is a constant vector.
12. Find the Directional derivative of $\phi = x^2 yz + 4xz^2 + xyz$ at $(1, 2, 3)$ in the direction of $2\mathbf{i} + \mathbf{j} - \mathbf{k}$.

SECTION– B ($5 \times 5 = 25$ marks)

Answer any FIVE questions

13. Find the value of $\int_0^{2a} \frac{x^3 \, dx}{\sqrt{(2ax - x^2)}}$.
14. Write the reduction formula for $\int \sin^n x \, dx$.
15. Find the Fourier series of $f(x) = x$ in $(0, 2\pi)$ of period 2π .
16. Find the Inverse Laplace transform of $\{\log(1 + \frac{w^2}{s^2})\}$.
17. Solve $\sqrt{p} + \sqrt{q} = 1$.
18. Find the Laplace transform of $\frac{1 - \cos t}{t}$.
19. If $\mathbf{r}$ is the position vector of the point $P(x, y, z)$ prove i) $\text{div } \mathbf{r} = 3$ ii) $\text{curl } \mathbf{r} = 0$.

SECTION– C ($3 \times 10 = 30$ marks)

Answer any THREE questions

20. If $f(m, n) = \int_0^{\frac{\pi}{2}} \cos^m x \cos nx \, dx$ p. t $f(m, n) = \frac{m}{m+n} f(m-1, n-1)$.
Prove that $f(n, n) = \frac{\pi}{2^{n+1}}$.
21. Find the fourier series of periodicity 2π for $f(x) = x^2$ in $(0, 2\pi)$.
Deduce $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$.
22. Solve $\mathbf{z}^2(\mathbf{p}^2 \mathbf{x}^2 + \mathbf{q}^2) = 1$.
23. Find the Laplace transform of $L\left[\frac{1 - \cos t}{t^2}\right]$ and $L\left[\frac{\sinh at}{t}\right]$
24. Prove that $\text{curl curl } \mathbf{F} = \text{grad div } \mathbf{F} - \nabla^2 \mathbf{F}$ and hence deduce that $\text{curl curl curl curl } \mathbf{F} = \nabla^4 \mathbf{F}$ if $\mathbf{F}$ is Solenoidal.

